

# Matrix Equation

E-Vex

$$\begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$$

Become:

$$\begin{cases} 3x - y = 5 \\ 2x + 4y = 8 \end{cases} \Rightarrow \begin{aligned} -y &= 5 - 3x \\ y &= \underline{\underline{-5 + 3x}} \end{aligned}$$

$$\Rightarrow 2x + 4(-5 + 3x) = 8$$

$$2x + (-20 + 12x) = 8$$

$$2x + 12x = 28$$

$$14x = 28$$

$$x = \frac{28}{14} \Rightarrow \boxed{x = 2}$$

$$\text{then: } 3(2) - y = 5$$

$$6 - y = 5$$

$$-y = -1$$

$$\Rightarrow \boxed{y = 1}$$

$$S: \{(2, 1)\}$$

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